

Volatility-of-volatility risk in the crude oil market

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Abstract

This paper examines the role of oil volatility-of-volatility (VOV) risk under a stochastic VOV framework. We show that oil VOV is a significant pricing factor in the cross-sectional delta-hedged gains constructed from oil options, and oil VOV also has predictive power for near-term delta-hedged option gains. Moreover, we show that the information contained in oil VOV is highly specific compared to its equity counterpart and other volatility-related measures, from the perspective of its predictability of future economic conditions. Our findings are robust to alternative VOV risk measures and forecasting horizons.

Keywords: Crude oil market; Delta-hedged gains; Jump risk; Pricing implications; Stochastic volatility-of-volatility risk

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1. Introduction

Despite the well-established strand of literature on the impact of stochastic volatility in financial asset pricing, from both theoretical and empirical perspectives (e.g., [Ang et al., 2006](#); [Campbell et al., 2018](#); [Cao and Han, 2013](#); [Heston, 1993](#); [Hull and White, 1987](#)), much less is known about the role of the uncertainty of volatility risk in asset pricing and investing decisions. In general, uncertainty of volatility that characterizes the distribution of time-varying volatility is empirically measured as the volatility-of-volatility (VOV). Intuitively, VOV also carries important information about the investment opportunity set, as shocks to the economy cause sudden increases in VOV of the market portfolio, which eventually affects investors' asset allocation decisions. This argument has also been theoretically supported by the dynamic pricing framework of asset returns proposed by [Huang et al. \(2019\)](#), which incorporates both time-varying volatility as well as the VOV risks. The analysis of VOV risk, however, is an underexplored topic, while being of great importance for researchers and practitioners.

Crude oil is used as a major input for production; fluctuations in the oil market have substantial impacts on the real economy (e.g., [Ferderer, 1996](#); [Elder and Serletis, 2010](#); [Hamdi et al., 2019](#); [Herrera et al., 2019](#)). It is further widely acknowledged that oil volatility has significant effects on financial markets, such as the stock market (e.g., [Feng et al., 2017](#)), the bond market (e.g., [Kang et al., 2014](#)), and the credit market (e.g., [Bouri et al., 2018](#)). The cross-sectional analysis of [Christoffersen and Pan \(2018\)](#) finds that stocks with higher exposure to oil volatility risk earn lower average returns, and vice versa. These studies highlight the fundamental significance of the crude oil market, and thus the uncertainty of oil market volatility, measured as oil VOV, can also be an important risk factor affecting consumption and

production decisions. In this paper, we aim to analyze whether aggregate VOV risk in the crude oil market is priced by the delta-hedged gains of United States Oil Fund (USO) options and whether oil VOV contains information that is distinct from that of its equity counterpart.⁵

Our research is related to the growing body of asset pricing literature on VOV risk, whose economic implications have been less frequently examined until recently. [Bollerslev et al. \(2009\)](#) incorporate stochastic VOV risk into the long-run risk model of [Bansal and Yaron \(2004\)](#) and they show that VOV risk drives the time-varying variance risk premium and contributes to the prediction of future equity index market returns. More recently, [Chen et al. \(2017\)](#) confirm the important role of VOV as a state variable that is significantly priced in the cross section of stock returns by developing a macroeconomic model that incorporates VOV risk as a pricing factor. In addition, several empirical papers have recently explored the cross-sectional implications of VOV risk in various settings, such as individual stocks (e.g., [Baltussen et al., 2018](#)), the equity index (e.g., [Hollstein and Prokopczuk, 2018](#)), and hedge funds (e.g., [Agarwal et al., 2017](#)). More recently, a few studies have examined the relation between VOV risk and options. [Huang et al. \(2019\)](#) extend the analysis of VOV as a risk factor to the Standard & Poor's (S&P) 500 and Chicago Board Options Exchange (CBOE) Volatility Index (VIX) option markets. [Cao et al. \(2018\)](#) investigate the cross-sectional effects of VOV risk in future delta-hedged equity index option returns. [Ruan \(2020\)](#) focuses on the significantly negative relation between VOV risk and the cross section of equity option returns. However, so far, the role of VOV in the crude oil market remains unknown.

⁵ The USO is one of the largest and most liquid crude oil exchange-traded funds (ETFs).

Our analysis extends the earlier discussions to the crude oil market by incorporating its VOV, while most previous work is confined to stochastic volatility. For instance, [Chiang et al. \(2015\)](#) show that a latent stochastic volatility factor is significantly related to key macro variables and is thus a significant pricing factor. [Prokopczuk et al. \(2017\)](#) observe a negative oil volatility risk premium by looking at the average of a synthetic variance swap rate constructed from oil futures options. Even though recent studies have identified the important role of VOV risk in the equity markets, the question of whether VOV risk retains the same significance in the crude oil market has not yet been explored.

To test our hypothesis, we first construct the VOV measure using the exponentially weighted moving average (EWMA) model with the CBOE Crude Oil Volatility Index (OVX).⁶ Next, we compare our findings with equity index volatility and VOV risks. The equity market volatility index and VOV index are proxied by VIX and VVIX, respectively.⁷ The correlation is much lower between the oil and equity index VOV than between oil and equity market volatility, indicating that oil and equity VOV risks contain distinct information than their volatility counterparts.

Our research is the first attempt to investigate whether uncertainty about oil market volatility is priced in the cross section of oil option returns. [Bakshi et al. \(2003\)](#) conduct a similar study on whether volatility risk is cross-sectionally priced in delta-hedged gains of S&P 500 options. [Huang et al. \(2019\)](#) analyze the equity market VOV and option returns associated with

⁶ The OVX, or “oil VIX,” measures the volatility of the crude oil market over the next one month.

⁷ The VIX is a forward measure of the stock market’s expected volatility over the next month. The VVIX represents the expected volatility in the one-month-ahead price of the VIX.

the S&P 500 and VIX, and they also provide a testable model in which the expected delta-hedged gains can be expressed as a sum of the market price of risk and compensations for volatility and VOV risks. [Hollstein and Prokopczuk \(2018\)](#) also confirm that the aggregate VOV risk is priced in the cross section of equity returns and has negative predictive power for the future volatility and VOV. We adopt the methodology of [Huang et al. \(2019\)](#) to test whether oil volatility and VOV risks are cross-sectionally priced in delta-hedged USO option returns.⁸

We further investigate the predictive power of VOV risk in the crude oil market. To do so, we run a set of time-series regressions controlling for various jump measures. We find that both oil market volatility and VOV risks significantly and negatively predict the one-month-ahead delta-hedged gains of USO options, consistent with previous studies of equity index and VIX markets (e.g., [Huang et al., 2019](#)). We also conduct several robustness tests by adopting a different construction method for VOV risk and considering a one-week forecasting horizon by using weekly USO options.

We further identify the specific information content on oil VOV from the perspective of its predictability for future economic. More specifically, we find that higher oil VOV leads to lower personal consumption and higher financial stress level by investigating forecasting ability of oil VOV on one-month-ahead growth rate of real personal consumption expenditure (RPCE) and Kansas City Financial Stress Index (KCFSI), respectively. In sum, oil VOV contains

⁸ Because ETFs and equity have a similar structure, USO options can provide equity option-like features, which are simpler than options on West Texas Intermediate (WTI) crude oil futures.

unique market information that cannot be covered by its equity counterpart and other volatility-related measures.

Our paper contributes to the literature in several respects. First, we construct the oil market's VOV measure using the EWMA model with the OVX index. Second, we examine how oil VOV risk is priced in delta-hedged option gains. Studies related to our analysis include [Baltussen et al. \(2018\)](#) and [Agarwal et al. \(2017\)](#) who find that VOV is an important factor in explaining cross-sectional returns of stocks and hedge funds, respectively. Third, we find that oil VOV negatively predicts future delta-hedged gains of USO options, consistent with the findings of [Huang et al. \(2019\)](#) in the equity index market. Fourth, the specific information content of oil VOV can be demonstrated by its predictability for future economic conditions. To the best of our knowledge, our work represents the first effort to comprehensively examine the VOV risk in the crude oil market by providing both cross-sectional and time-series evidence.

The remainder of this paper proceeds as follows. Section 2 describes the data and key variables. Section 3 describes the empirical analysis, including the results of testing the market price of oil VOV risk and its specific information content. Section 4 reports a set of robustness tests in which we consider jump risks, alternative measure of oil VOV, and one-week USO delta-hedged gains. Finally, Section 5 concludes the paper.

2. Data and Key Variables

2.1. Data Information

The empirical analysis spans from July 2010 to June 2018. We use the USO option data obtained from the Thomson Reuters Tick History of Eikon to construct the delta-hedged gains.⁹ Before our calculation, we filter out option data with incomplete or incorrect information. Specifically, we remove options with a zero close bid, a close ask greater than the close bid, prices violating the standard no-arbitrage condition, or a Black-Scholes-implied volatility greater than 100% or less than 1%. The following analysis mainly focuses on delta-hedged gains obtained from one-month options. Weekly USO options were launched in July 2010, and we also construct delta-hedged gains using weekly options for a robustness check to compare with their one-month results. Moreover, we use the five-minute intraday prices of the USO exchange-traded funds (ETFs) obtained from Thomson Reuters DataScope Select to calculate the model-free jump measures proposed by [Barndorff-Nielsen et al. \(2004\)](#) and [Barndorff-Nielsen and Shephard \(2006\)](#). The risk-free rates are proxied by the London Interbank Offered Rate, obtained from Federal Reserve Economic Data (FRED) offered by the Federal Reserve Bank of St. Louis.¹⁰

2.2. Key Variable Definitions

2.2.1. Volatility-of-Volatility (VOV) Measures

⁹ Compared to options written on crude oil futures, ETF option data have the advantage of an equity option-like structure, which is much simpler.

¹⁰ <https://www.stlouisfed.org>.

We adopt the OVX as a proxy for crude oil market volatility, which measures the 30-day-ahead volatility of the market.¹¹ We then construct the oil VOV measure, denoted by $vo v_t^2$, by using the EWMA model with the OVX index:¹²

$$vo v_t^2 = \lambda vo v_{t-1}^2 + (1 - \lambda)u_{t-1}^2, \quad (1)$$

where u_t is the logarithm of the gross return of the OVX (i.e., $\ln\left(\frac{OVX_t}{OVX_{t-1}}\right)$), $vo v_t$ is the conditional volatility of the gross OVX return, and λ is the degree of the weighting decrease, which is set at 0.94 here.

2.2.2. Delta-Hedged Option Gains

A delta-hedged option portfolio consists of longing an option and shorting delta units, as implied by the option, of its underlying stock, with net income invested at the risk-free rate. It is worthwhile to note that the gains of such a portfolio are thus insensitive to changes in the prices of the underlying stock.

Let $C(t, \tau, K)$ denote the price at time t of a call option with time to maturity τ and strike price K and let the corresponding option delta be represented by $\Delta(t, \tau, K)$ (i.e., $\Delta(t, \tau, K) \equiv \frac{\partial C(t, \tau, K)}{\partial S_t}$). Then, the delta-hedged gains, denoted by $\pi_{t, t+\tau}$, can be expressed as

$$\pi_{t, t+\tau} = C_{t+\tau} - C_t - \int_t^{t+\tau} \Delta_u dS_u - \int_t^{t+\tau} r_u (C_u - \Delta_u S_u) du, \quad (2)$$

¹¹ The OVX is the oil VIX. The construction method of the OVX is similar to that for the VIX.

¹² The EWMA model is widely used in practice because of its simplicity and less restrictive data requirements.

where S_t denotes the price at time t of the underlying stock and r_t denotes the risk-free rate at time t . For compactness, we denote $C(t, \tau, K)$ and $\Delta(t, \tau, K)$ as C_t and Δ_t , respectively.

2.3. Summary Statistics

2.3.1. Volatility-Related Variables

Table 1 reports the descriptive statistics for crude oil market volatility (*vol*), crude oil market VOV (*vov*), equity market volatility (VIX), and equity market VOV (VVIX).¹³ Notably, *vov* exhibits much greater skewness and kurtosis than the other variables, suggesting that oil VOV is highly asymmetric.

[Insert Table 1 here]

Table 2 reports the cross-sectional correlations between the variables. The correlation between oil market volatility (*vol*) and VOV (*vov*) is 0.250, suggesting that limited co-movement between volatility and VOV exists in the crude oil market. This result is also consistent with the time-series plots of *vol* and *vov*, as depicted in Figure 1. Interestingly, the spikes exhibited by *vol* and *vov* reveal markedly different patterns. Moreover, the correlation between oil market volatility (*vol*) and equity market volatility (VIX) is 0.493, much higher than the correlation between oil market VOV (*vov*) and equity market VOV (VVIX), which is

¹³ We also incorporate equity market volatility and VOV risks here for comparison with their crude oil market counterparts.

around 0.169. This result highlights the fact that volatility and VOV risks in the crude oil and equity markets contain distinct information.

[Insert Table 2 here]

[Insert Figure 1 here]

2.3.2. Statistical Properties of Delta-Hedged Gains

We next consider the discrete-time counterpart of Equation (2), with daily rebalancing:

$$\pi_{t,t+\tau} = C_{t+\tau} - C_t - \sum_{n=0}^{N-1} \Delta_{t_n} (S_{t_{n+1}} - S_{t_n}) - \sum_{n=0}^{N-1} r_{t_n}^f (C_t - \Delta_{t_n} S_{t_n}) \frac{\tau}{N}, \quad (3)$$

where Δ_{t_n} is the option delta on day t_n , implied by the model of Black and Scholes (1973), and N is the total number of trading days from time t to $t + \tau$.

Table 3 reports the descriptive statistics for unscaled delta-hedged gains (i.e., $\pi_{t,t+\tau}$) and scaled delta-hedged gains (i.e., $\frac{\pi_{t,t+\tau}}{S_t}$) constructed on one-month USO call and put options, grouped by moneyness (i.e., $\frac{S_t e^{r\tau}}{K}$). On average, both unscaled and scaled delta-hedged gains are significantly negative over all ranges of moneyness, suggesting that the strategies lose money, which is consistent with previous findings for the equity markets (e.g., [Bakshi and Kapadia, 2003](#); [Cao and Han, 2013](#); [Carr and Wu, 2009](#)).

[Insert Table 3 here]

We observe that, in most cases, the magnitudes of the delta-hedged gains decrease when going deeper in or out of the money. Considering put options with moneyness $m \in [0.900, 0.925]$ versus put options with moneyness $m \in [0.975, 1.000]$, we can observe that the averages of the unscaled delta-hedged gains are $-\$0.068$ versus $-\$0.071$, and the averages of the scaled delta-hedged gains are -0.19% versus -0.21% . The curve for vega (i.e., $\frac{\partial C_t}{\partial \sigma_t}$) is concave, that is, it peaks for at-the-money (ATM) strikes and tends to decrease for away-from-the-money strikes, as depicted in Figure 2. This suggests that the impact of volatility risk decreases for away-from-the-money options. However, we also observe that the unscaled delta-hedged gains for put options with moneyness $m \in [0.925, 0.950]$ are $-\$0.076$, and the scaled delta-hedged gains are -0.20% , on average, where the former is greater than and the latter is similar to their ATM counterparts. This result is inconsistent with the decreasing magnitude of vega for away-from-the-money options. The curve of volga (i.e., $\frac{\partial^2 C_t}{\partial \sigma_t^2}$) is convex around the near-the-money range, as presented by Figure 2, which reaches its lowest point for ATM strikes, and tends to increase when moving away from moneyness equal to one. This result suggests increasing VOV risk for away-from-the-money options.

[Insert Figure 2 here]

We further find that the losses of delta-hedged portfolios increase with extended hedging horizons. This is easily observed from the delta-hedged gains constructed on one-week USO options, reported in Table 4. For example, for the call option groups with moneyness $m \in [0.975, 1.000]$, the average of unscaled delta-hedged gains (i.e., $\pi_{t,t+\tau}$) is $-\$0.031$ for the one-week and $-\$0.056$ for one-month options, respectively. We observe similar patterns for options

in other moneyness groups. Thus, delta-hedged gains tend to increase with longer hedging horizons.

[Insert Table 4 here]

Overall, the delta-hedged gains are negative for all options in the crude oil market, and the non-monotonic changes of the delta-hedged gains with respect to moneyness indicate the possible impacts of both volatility and VOV risks in the crude oil market.

3. Empirical Analysis

Previous studies find that returns of the delta-hedged strategy in the equity markets are negative (e.g., [Bakshi and Kapadia, 2003](#); [Carr and Wu, 2009](#)). [Bakshi and Kapadia \(2003\)](#) confirm the significant impact of volatility risk on negative delta-hedged gains in the equity index market. Recent work by [Huang et al. \(2019\)](#) shows that in the equity index market VOV risk also plays a significant role in pricing delta-hedged gains. However, the question of how the delta-hedged strategy performs in the crude oil market and whether the corresponding volatility and VOV risks significantly affect the portfolio gains remains unexplored.

3.1. Delta-Hedged Gains and Stochastic Volatility-of-Volatility Risk

3.1.1. Stochastic Volatility-of-Volatility Model

Following the framework of [Huang et al. \(2019\)](#), which incorporates time-varying volatility and VOV risks, the dynamics of the underlying price S_t under the physical measure P is such that

$$\frac{dS_t}{S_t} = \mu(S_t, v_t, \eta_t)dt + \sqrt{v_t}d\omega_t^1, \quad (4)$$

$$dv_t = \theta(v_t)dt + \sqrt{\eta_t}d\omega_t^2, \quad (5)$$

$$d\eta_t = \gamma(\eta_t)dt + \phi\sqrt{\eta_t}d\omega_t^3, \quad (6)$$

where $(\omega_t^1, \omega_t^2, \omega_t^3)_{t \geq 0}$ is a three-dimensional Brownian motion with the correlation coefficient $\text{corr}(d\omega_t^i, d\omega_t^j) = \rho_{ij}$, for all $i \neq j$, v_t denotes the stochastic variance of immediate returns, and η_t denotes the variance of innovation in v_t , which follows an autonomous stochastic process.

Based on this framework, the expected value of the delta-hedged gain (i.e., $\pi_{t,t+\tau}$) depends on the risk premiums of volatility and VOV risks, as shown by

$$E_t[\pi_{t,t+\tau}] = \int_t^{t+\tau} E_t \left[\lambda_u^v \frac{\partial C_u}{\partial v_u} \right] du + \int_t^{t+\tau} E_t \left[\lambda_u^\eta \frac{\partial C_u}{\partial \eta_u} \right] du, \quad (7)$$

where λ_t^v and λ_t^η denote the compensation for stochastic volatility and VOV risks, respectively.

Moreover, $\frac{\partial C_u}{\partial v_u}$ and $\frac{\partial C_u}{\partial \eta_u}$ represent the vega and volga of the call option, respectively.

Furthermore, assuming that the risk premiums (i.e., λ_t^v and λ_t^η) have a linear structure, the scaled expected delta-hedged gain is linear with respect to the fundamental factors:

$$\frac{E_t[\pi_{t,t+\tau}]}{S_t} = \lambda^v \beta_t^v v_t + \lambda^\eta \beta_t^\eta \eta_t, \quad (8)$$

where β_t^v and β_t^η represent the options' exposure to volatility risk and VOV risk, respectively. For more details about the expressions β_t^v and β_t^η and corresponding derivations, see Appendix A.1.

Overall, under the stochastic volatility and VOV pricing framework proposed by [Huang et al. \(2019\)](#), the expected value of the delta-hedged gains compensates for both volatility and VOV risks. Moreover, if restrictions in the linear structure are placed on the risk premiums, the scaled delta-hedged gains are also linear with respect to volatility and VOV risks. We conduct corresponding empirical analysis in the following section.

3.1.2. Cross-Sectional Analysis

The stochastic VOV model implies that options with higher exposure to volatility and VOV risks should have a higher negative payoff. To empirically verify this justification, we follow the extant literature (e.g., [Bakshi and Kapadia, 2003](#); [Huang et al., 2018](#)) and employ the option greeks (i.e., vega and volga) implied by the model of [Black and Scholes \(1973\)](#) to proxy for option volatility and VOV risk betas. Then, we adopt the following econometric specification to check the cross-sectional implications of delta-hedged option gains on volatility and VOV risks:

$$Gains_{t,t+\tau}^i = \Phi_1 Vega_t^i + \Phi_2 Volga_t^i + \gamma_t + \epsilon_{t,t+\tau}^i, \quad (9)$$

where we define scaled delta-hedged gains as $Gains_{t,t+\tau}^i \equiv \frac{\pi_{t,t+\tau}^i}{S_t}$. We also denote the option vega and volga implied by [Black and Scholes \(1973\)](#) as $Vega_t^i$ and $Volga_t^i$, respectively. Let i

represent the moneyness range ($i = 1, 2, \dots, I$) corresponding to the option groups shown in Table 3. We use γ_t to denote the time-fixed effects.

Table 5 reports the cross-sectional evidence from the regressions of the average scaled delta-hedged gains (i.e., $Gains^i$) on option volatility and VOV risk betas. The univariate estimate for vega is significantly negative, with a value of -0.058 and a t -statistic of -4.80. The negative volatility premiums in the crude oil market are consistent with those in previous findings (e.g., [Bakshi and Kapadia, 2003](#); [Bardgett et al., 2019](#); [Bollerslev et al., 2011](#)). The second row shows the univariate estimate for volga, which is insignificant. The third row presents the multivariate regression including both vega and volga. Both estimates are negative and statistically significant at the 1% level; the slope of vega is -0.119, and the corresponding t -statistic is -7.22; the slope of volga is -0.020, and the corresponding t -statistic is -5.38.¹⁴ The negative price of VOV risk in the crude oil market is consistent with that in the equity index market as reported by [Huang et al. \(2019\)](#). In sum, the cross-sectional evidence indicates that both volatility and VOV risks are negatively and significantly priced in the crude oil market.

[Insert Table 5 here]

3.1.3. Time-Series Analysis

Under the stochastic VOV model, we can see that time variations in the expected delta-hedged gains are driven by time-varying volatility and VOV risks. Thus, we adopt the following

¹⁴ We scale both slopes by 1,000 here.

econometric specification to examine the impacts of both risks on the time variations in scaled delta-hedged gains:

$$Gains_{t,t+\tau}^i = \Phi_0 + \Phi_1 vol_t^2 + \Phi_2 vov_t^2 + Gains_{t-\tau,t}^i + u^i + \epsilon_{t+\tau}^i, \quad (10)$$

where we include the moneyness group fixed effects u^i to account for heterogeneity in the sensitivity of options in various moneyness bins to the underlying volatility and VOV risks. vol_t represents oil market volatility, vov_t represents oil market VOV, and $Gains_{t-\tau,t}^i$ represents one-month-lagged delta-hedged gains, with i as the moneyness ranges of the USO options.

[Insert Table 6 here]

The results are reported in Table 6. The first column shows the results for the case in which vov_t^2 is a single predictor. It significantly predicts one-period-ahead delta-hedged gains, with a negative coefficient of -0.894 and a t -statistic of -7.83 , consistent with the results of cross-sectional analysis. The second column of Table 6 is a joint prediction model incorporating both vol_t^2 and vov_t^2 , and it shows that the coefficients of both are negative and highly significant. The findings for the crude oil market are very similar to those for the equity index market, as demonstrated by [Bakshi and Kapadia \(2003\)](#) and [Huang et al. \(2019\)](#). Overall, the time-series evidence also indicates that both volatility and VOV risks are negatively and significantly priced in the crude oil market.

3.2. The Information Contained in Oil Volatility-of-Volatility Risk

So far, we have confirmed that both oil volatility and VOV risks are significantly and negatively priced in the USO option market, and this section builds on previous research by [Huang et al. \(2019\)](#) that focuses on the relationship between equity volatility and VOV risks and delta-hedged S&P 500 option gains. Uncertainty about oil volatility could also affect financial market conditions and the real economy because economic decisions in a state with high oil market volatility are largely different from those in a state with low oil market volatility. In this section, we further explore whether the information content of oil VOV is distinct from its equity market counterpart and oil volatility, from the perspective of its predictability for future economic conditions.¹⁵

3.2.1. Economic Conditions: Consumption Growth and Financial Stress Level

Oil prices have significant impacts on the U.S. economy. Intuitively, oil prices influence production and manufacturing costs, which ultimately affects business and consumption. Meanwhile, because of the rapid development of new technology allowing extraction of oil and natural gas from shale deposits, the U.S. has become a major oil-producing economy, therefore, fluctuations in oil prices also affect banking and finance based on the changing financial needs of oil companies. In sum, oil prices affect the U.S. economy in two ways: the rapid development of the oil industry has positive impacts on banking and finance and lowers consumption and business costs. We thus analyze the predictive relationships between oil market VOV and future

¹⁵ It has been shown that the volatility risk of the oil and equity markets differ, and both are important pricing factors (e.g., [Christoffersen and Pan, 2018](#); [Gao et al., 2018](#)).

economic conditions, such as the growth rate of RPCE and the financial stress level proxied by KCFSI.¹⁶ Specifically, we run the following predictive regression:

$$y_{t+1} = \alpha + \beta_h^1 vov_t^2 + \beta_h^2 vol_t^2 + \beta_h^3 VVIX_t^2 + \beta_h^4 y_t + \gamma_h' x_t + \epsilon_t, \quad (11)$$

where y stands for the economic variables of interest (i.e., RPCE and KCFSI), vov represents the oil market VOV, vol is the oil market volatility, $VVIX$ is the equity market VOV, and x is the vector of fundamental economic variables, including the term spread (TS), the default spread (DS), and the log dividend-price ratio (DP).

The results are reported in Table 7. Oil VOV (vov) negatively forecasts the growth rate of consumption, indicating that it contains different information from other volatility-related variables, such as oil volatility (vol) and equity VOV ($VVIX$). These results imply that oil VOV could be closely related to the real economy, and fluctuations in oil volatility have a negative and significant effect on consumption growth. One possible reason for this is that higher fluctuations in oil volatility may lower consumption of durable goods ([Elder and Serletis, 2010](#)). Moreover, higher uncertainty in oil volatility may affect consumer spending because it raises uncertainty about employment, and thus people increase their precautionary savings ([Edelstein and Kilian, 2009](#)). Additionally, the influence of commodity markets on the global financial market has increased because of the financialization of commodity futures markets starting in 2004-2005. These motivate us to investigate the predictive power of oil VOV on the financial market stress

¹⁶ The Kansas City Financial Stress Index (KCFSI) is a monthly measure of stress in the U.S. financial system based on 11 financial market variables and can be obtained from Federal Reserve Economic Data (FRED).

level.¹⁷ We find that both oil and equity VOV significantly and positively predict one-month-ahead KCFSI with corresponding t-statistics of 2.58 and 4.06, respectively. Our finding is consistent with that of [Nazlioglu et al. \(2015\)](#), that downturns in the oil market tend to depress economic activity, which negatively affects banking and finance and subsequently increases financial stress. In fact, previous research such as [Nazlioglu et al. \(2015\)](#) has provided mixed evidence on the relationship between the oil market and the financial market, and our findings contribute to this strand of the literature.

Overall, oil VOV contains specific predictive information on economic conditions, in that higher oil VOV can significantly predict a lower growth rate in personal consumption and higher financial stress in the near term.

[Insert Table 7 here]

4. Robustness

4.1. Robustness Check: Control for Jump Risks

Under the stochastic VOV model, as specified by Equations (4)-(6), jump risk is not considered; however, as demonstrated in previous research, adding jumps can improve the performance of pricing models (e.g., [Bakshi et al., 1997](#); [Bates, 1996](#); [Duffie et al., 2000](#)). In this section, we introduce jump risk to the model, which considers the impacts of stochastic volatility risk,

¹⁷ For further discussion about the effects of the financialization of commodity futures markets, see [Tang and Xiong \(2012\)](#).

stochastic VOV risk, and jump risk. We can thus further investigate whether the pricing role of oil VOV is affected by jumps.

4.1.1. Stochastic Volatility-of-Volatility Model with Jumps

As shown by [Huang et al. \(2019\)](#), if jump risk is considered, the dynamics of underlying price S_t under the physical measure P is

$$\frac{dS_t}{S_t} = \mu(S_t, v_t, \eta_t)dt + \sqrt{v_t}d\omega_t^1 + (e^x - 1)dq_t - \mu_J \Lambda_J \sqrt{v_t}dt, \quad (13)$$

$$dv_t = \theta(v_t)dt + \sqrt{\eta_t}d\omega_t^2, \quad (14)$$

$$d\eta_t = \gamma(\eta_t)dt + \phi\sqrt{\eta_t}d\omega_t^3. \quad (15)$$

The jump term follows a Poisson process with volatility-independent intensity $\Lambda_J \sqrt{v_t}$. We use x to denote the jump size and q to represent its corresponding physical density. We further assume that the mean of $e^x - 1$ is μ_J , so the compensator is $\mu_J \Lambda_J \sqrt{v_t}dt$.

Therefore, the expected delta-hedged gain (i.e., $\pi_{t,t+\tau}$) under physical measure P compensates for stochastic volatility, VOV, and jump risks, as shown by

$$\begin{aligned} E_t[\pi_{t,t+\tau}] = & \int_t^{t+\tau} E_t \left[\lambda_u^v \frac{\partial C_u}{\partial v_u} \right] du + \int_t^{t+\tau} E_t \left[\lambda_u^\eta \frac{\partial C_u}{\partial \eta_u} \right] du + \mu_J^* \Lambda_J \int_t^{t+\tau} E_t \left[\sqrt{v_u} S_u \frac{\partial C_u}{\partial S_u} \right] du \\ & - \Lambda_J \int_t^{t+\tau} \sqrt{v_u} \left[\int_{-\infty}^{\infty} C(S_u e^x) q(x)^* dx - \int_{-\infty}^{\infty} C(S_u e^x) q(x) dx \right] du. \end{aligned} \quad (16)$$

In sum, if stochastic volatility, VOV, and jump risks are considered in the dynamics of underlying asset S_t , the expected delta-hedged gains compensate for all three risks. For more details about the derivation, see Appendix A.2.

4.1.2. Jump Risk Measures

If jump risk is considered, as shown in Equation (16), the delta-hedged gain could also be attributed to the large, discontinuous jumps in the underlying prices, apart from compensation for the VOV risk. To address concerns about whether the effect of VOV risk is robust to the presence of jump risk, we adopt two measures of jump risk.

The first jump measure is risk-neutral skewness and kurtosis, because jump risk can be proxied by the skewness and kurtosis of the underlying return distribution (e.g., [Bakshi et al., 2003](#); [Bates, 2000](#); [Jackwerth and Rubinstein, 1996](#)), in which the former is closely related to jump size, and the latter is closely related to jump intensity. We construct risk-neutral skewness and kurtosis using the model-free approach of [Bakshi et al. \(2003\)](#). Specifically, the higher-order risk-neutral moments can be spanned by a set of out-of-the-money calls and puts,

$$skew_{t,t+\tau} = \frac{e^{r\tau}W_{t,t+\tau} - 3\mu_{t,t+\tau}e^{r\tau}V_{t,t+\tau} + 2\mu_{t,t+\tau}^3}{[e^{r\tau}V_{t,t+\tau} - \mu_{t,t+\tau}^2]^{3/2}}, \quad (17)$$

$$kurt_{t,t+\tau} = \frac{e^{r\tau}X_{t,t+\tau} - 4\mu_{t,t+\tau}e^{r\tau}W_{t,t+\tau} + 6e\mu_{t,t+\tau}^2V_{t,t+\tau} - 3\mu_{t,t+\tau}^4}{[e^{r\tau}V_{t,t+\tau} - \mu_{t,t+\tau}^2]^2}, \quad (18)$$

where $V_{t,t+\tau}$, $W_{t,t+\tau}$, and $X_{t,t+\tau}$ are based on the prices of volatility and cubic and quartic contracts, respectively, and $\mu_{t,t+\tau}$ is a function of them. Detailed expressions are in Appendix Section A.3.

The second jump measure is constructed by applying the model-free methodology of [Barndorff-Nielsen et al. \(2004\)](#) and [Barndorff-Nielsen and Shephard \(2006\)](#) to high-frequency

USO prices. According to [Barndorff-Nielsen et al. \(2004\)](#), the realized variance, the sum of squared intraday returns, is a consistent estimator for integrated volatility when there are no jumps:

$$RV_t = \sum_{i=1}^{M_t} r_{t,i}^2, \quad (19)$$

where M_t is the total number of observations, and $r_{t,i}$ is the i th five-minute return on trading day t . [Barndorff-Nielsen et al. \(2004\)](#), [Barndorff-Nielsen and Shephard \(2006\)](#), and [Huang and Tauchen \(2005\)](#) further show that the bipower variation, the sum of the products of adjacent absolute intraday returns, is a consistent estimator of integrated volatility, even in the presence of jumps:

$$BV_t = \frac{\pi}{2} \frac{M}{M-1} \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}|. \quad (20)$$

Therefore, the jump measure can be constructed based on the realized variance and bipower variation:

$$J_t = \text{sign}(r_t) \times \sqrt{(RV_t - BV_t) \times I(ZJ_t \geq \Phi_\alpha^{-1})}, \quad (21)$$

where Φ_α^{-1} denotes the inverse cumulative distribution function of the standard normal distribution. We set $\alpha = 99.9\%$, and $I(\cdot)$ is an indicator function. The specific expression for ZJ_t is in Appendix Section A.4.

Furthermore, the model-free jump risk measure (J) can be grouped into positive and negative jumps, based on the sign of the market return on the day:

$$J_t^+ = I(r_t \geq 0)J_t, \quad (22)$$

$$J_t^- = I(r_t < 0)J_t, \quad (23)$$

where r_t denotes the daily return on day t .

4.1.3. Empirical Analysis

So far, we have identified the effects of oil volatility and VOV risks on the delta-hedged payoff of USO options. The losses on the delta-hedged portfolios can also be attributed to the fear of market crashes, namely, jump risk, as demonstrated in the model of stochastic VOV with jumps in Section 4.1.1. In this section, we empirically verify the joint effects of volatility, VOV, and jump risks on scaled delta-hedged gains by including the jump risk measures. Specifically, we consider two sets of jump measures, described in Section 4.1.2.

The first set of jump measures considered is risk-neutral skewness and kurtosis extracted from the cross section of USO option prices, using the model-free approach of [Bakshi et al. \(2003\)](#). The econometric specification is as follows:

$$Gains_{t,t+\tau}^i = \Phi_0 + \Phi_1 vol_t + \Phi_2 vov_t + \Phi_3 skew_t + \Phi_4 kurt_t + Gains_{t-\tau,t}^i + u^i + \epsilon_{t+\tau}^i, \quad (24)$$

where $skew_t$ and $kurt_t$ denote risk-neutral skewness and kurtosis, respectively, on trading day t .

Columns (3) to (5) in Table 6 display the results after controlling for risk-neutral skewness ($skew$) and kurtosis ($kurt$). The predictive power of oil volatility (vol) and VOV (vov) risks is not affected by the risk-neutral jump measures, and the corresponding slopes Φ_1 and Φ_2 remain unchanged. Moreover, regarding the predictability of risk-neutral jumps, greater jump fear risk predicts more negative delta-hedged gains in the future: $kurt$ significantly and negatively predicts future gains, whether alone or combined with $skew$. This finding is

consistent with similar studies on the equity index market (e.g., [Bakshi et al., 2003](#); [Huang et al., 2018](#)).

The second set of jump measures is constructed from the five-minute intraday USO data, based on Equation (21). We further divide the jump measure into positive and negative components based on market direction, as shown in Equations (22) and (23). Thus, we consider the following two sets of regressions:

$$Gains_{t,t+\tau}^i = \Phi_0 + \Phi_1 vol_t + \Phi_2 vov_t + \Phi_3 J_t + \Phi_4 Gains_{t-\tau,t}^i + u^i + \epsilon_{t+\tau}^i, \quad (25)$$

$$Gains_{t,t+\tau}^i = \Phi_0 + \Phi_1 vol_t + \Phi_2 vov_t + \Phi_3 J_t^+ + \Phi_4 J_t^- + \Phi_5 Gains_{t-\tau,t}^i + u^i + \epsilon_{t+\tau}^i, \quad (26)$$

where J_t represents the jump measure for trading day t , and J_t^+ and J_t^- stand for positive and negative jumps, respectively.

We control for the realized jump measures, both decomposed and undecomposed, and obtain the results reported in Columns (6) and (7) in Table 6. The variables vol and vov still strongly and negatively predict future delta-hedged option gains, even after the control variables are included. As when the risk-neutral jumps are controlled for, the coefficients of realized jump measures (i.e., J , J^+ , and J^-) are highly significant. Overall, the predictability of oil VOV risk is robust after controlling for jump risk measures.

4.2. Robustness Check: Alternative Volatility-of-Volatility Measure

We construct a second measure of oil VOV following [Baltussen et al. \(2018\)](#). Specifically, we scale the standard deviation of oil market volatility, which is still proxied by OVX, with the average volatility over the past month, as follows:

$$vov_t^B = \frac{\sqrt{\frac{1}{22} \sum_{j=t-21}^t (vol_t - \overline{vol}_t)^2}}{\overline{vol}_t}, \quad (27)$$

where vov^B is the alternative measure of oil VOV, vol is oil volatility, and $\overline{vol}_t = \sum_{j=t-21}^t vol_j$.

[Insert Table 8 here]

The results are displayed in Table 8. As in our previous findings, the univariate regression of delta-hedged USO option gains on oil VOV (vov^B) generates a highly significant and negative coefficient. Moreover, in the multivariate regressions that add oil volatility and jump measures as control variables, we find that vov^B remains highly significant and negative. This result suggests that oil VOV is significantly priced in the delta-hedged gains of USO options even with an alternative measure.

4.3. Impacts of Maturities: Weekly Options

Weekly options, or “weeklies,” were first introduced by the CBOE in October 2005. They have an expiration of about one week, whereas traditional options normally have an expiration of months or years. The addition of weekly options produces benefits such as greater trading flexibility and more precise timing; the trading in weeklies has grown rapidly.¹⁸ USO weekly options were introduced in July 2010. Therefore, we also construct delta-hedged gains using weekly USO options. We then conduct an analysis similar to that for one-month delta-hedged gains to investigate the impacts on them of volatility, VOV, and jump risks.

¹⁸ For example, according to [Andersen et al. \(2017\)](#), the trading share of S&P 500 weeklies has reached 40% of the total trading volume of S&P 500 options.

Using USO weeklies has two main advantages. First, as [Andersen et al. \(2017\)](#) state, the pricing of short-dated ATM options depends mainly on its diffusive component, whereas deep OTM options mainly reflect the characteristics of the risk-neutral jump process. The use of weekly ATM options can be an effective way to control for the effects of jump risk and to focus primarily on diffusive risks. Second, the use of weekly options can increase the sample size and complement our main results based on the use of monthly options.

[Insert Table 9 here]

Columns (1) to (3) in Table 9 show our results based on one-week delta-hedge option gains. The variables *vol* and *vov* remain significant and have negative predictive power for one-week-ahead delta-hedged option gains. Specifically, the *t*-statistic for the coefficient of *vov* is -5.01 for predicting one-week option gains, whereas the corresponding *t*-statistic is -3.26 for predicting one-month option gains. We also consider model specifications with jump risk proxies. Even after controlling for jump risk, we find that the coefficient of *vov* is still negative and significant; however, the coefficient of *vol* is no longer significant. Consistent with [Andersen et al. \(2017\)](#), where the effect of jump risk is lower for weekly ATM options, the predictive power of jump risk proxies is weaker than that associated with one-month delta-hedged option returns. Specifically, the coefficients of risk-neutral skewness (*skew*), risk-neutral kurtosis (*kurt*), and downside realized jump risk (J) are not significant, whereas those associated with realized jumps (J) and upside realized jumps (J^+) are statistically significant.

5. Conclusion

Based on the theoretical framework featuring stochastic VOV risk proposed by Huang et al. (2019), we provide a detailed empirical analysis on oil VOV risk. We demonstrate that oil VOV is a risk factor significantly priced in the cross section of USO option delta-hedged gains, beyond oil volatility risk. This finding is consistent with the descriptive statistics indicating: (1) the negative value of average delta-hedged USO option gains and their nonlinear pattern (average delta-hedged USO option gains) across moneyness suggest that investors need to be compensated for oil volatility and VOV risks, and (2) the low correlation between oil volatility and oil VOV, with a value of 0.250, indicates that the two risk factors contain different information.

We further explore the specific information content of oil VOV, in comparison with its equity counterpart and other volatility-related measures. First, we investigate the relationship between oil VOV and U.S. economic conditions, given the U.S. has recently become an oil economy. By focusing on the predictability of oil VOV on the one-month-ahead consumption growth rate and financial stress level, we find that oil VOV negatively predicts the personal consumption growth rate and positively predicts the financial stress level in the near term.

The predictive power of oil VOV remains significant after several robustness tests are performed. First, jump risk measures, such as risk-neutral skewness, risk-neutral kurtosis, and realized jumps extracted from high-frequency USO data, are incorporated into our analysis. Second, we adopt the alternative methodology proposed by [Baltussen et al. \(2018\)](#) and construct an oil VOV risk measure; we find that the new measure remains significant in predicting one-period-ahead delta-hedged gains. Finally, we change the forecasting horizon to one week by

using USO weekly option delta-hedged gains, and it complements our main findings which are based on monthly options.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from six different sources: Thomson Reuters Tick History of Eikon, Thomson Reuters DataScope Select, Thomson Reuters Datastream, Federal Reserve Economic Data (FRED), CBOE website (<https://www.cboe.com>), and Amit Goyal's website (<http://www.hec.unil.ch/agoyal>). Restrictions apply to the availability of data from Thomson Reuters Tick History of Eikon, Thomson Reuters DataScope Select, and Thomson Reuters Datastream, which were used under license for this study.

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- USO ETF data; Not publicly available; data can be obtained from Thomson Reuters DataScope Select (<https://www.refinitiv.com/en/products/datascope-select-data-delivery>) by paying a data license fee.
- USO option data; Not publicly available; data can be obtained from Thomson Reuters Tick History of Eikon (<https://www.refinitiv.com/en/financial-data/market-data/tick-history>) by paying a data license fee.

Appendix

A.1. Stochastic Volatility-of-Volatility Model

The pricing framework of [Huang et al. \(2019\)](#) features the dynamics of underlying returns by incorporating time-varying volatility and VOV risks. First, we assume that, under the physical measure P , the dynamics of the underlying price S_t are such that

$$\frac{dS_t}{S_t} = \mu(S_t, v_t, \eta_t)dt + \sqrt{v_t}d\omega_t^1, \quad (A1.1)$$

$$dv_t = \theta(v_t)dt + \sqrt{\eta_t}d\omega_t^2, \quad (A1.2)$$

$$d\eta_t = \gamma(\eta_t)dt + \phi\sqrt{\eta_t}d\omega_t^3, \quad (A1.3)$$

where $(\omega_t^1, \omega_t^2, \omega_t^3)_{t \geq 0}$ is a three-dimensional Brownian motion with correlation coefficient $\text{corr}(d\omega_t^i, d\omega_t^j) = \rho_{ij}$, for all $i \neq j$. v_t denotes the stochastic variance in immediate returns, and η_t denotes the variance of innovation in v_t .

Correspondingly, under the risk-neutral probability measure Q , the dynamics of the underlying price S_t are

$$\frac{dS_t}{S_t} = rdt + \sqrt{v_t}d\widetilde{\omega}_t^1, \quad (A1.4)$$

$$dv_t = (\theta(v_t) - \lambda_t^v)dt + \sqrt{\eta_t}d\widetilde{\omega}_t^2, \quad (A1.5)$$

$$d\eta_t = (\gamma(\eta_t) - \lambda_t^\eta)dt + \phi\sqrt{\eta_t}d\widetilde{\omega}_t^3, \quad (A1.6)$$

where $(\widetilde{\omega}_t^1, \widetilde{\omega}_t^2, \widetilde{\omega}_t^3)_{t \geq 0}$ represents the corresponding three-dimensional Brownian motion under the risk-neutral probability measure Q . Moreover, λ_t^v represents compensation for stochastic variance and λ_t^η represents compensation for the stochastic variance in the variance.

Consider a call option written on the underlying asset S_t . By Itô's lemma, its price at time $t + \tau$ can be expressed as

$$C_{t+\tau} = C_t + \int_t^{t+\tau} \frac{\partial C_u}{\partial S_u} dS_u + \int_t^{t+\tau} \frac{\partial C_u}{\partial v_u} dv_u + \int_t^{t+\tau} \frac{\partial C_u}{\partial \eta_u} d\eta_u + \int_t^{t+\tau} b_u du, \quad (A1.7)$$

where b_t is defined as

$$\begin{aligned} b_t = & \frac{\partial C_t}{\partial t} + \frac{1}{2} v_t S_t^2 \frac{\partial^2 C_t}{\partial S_t^2} + \frac{1}{2} \eta_t \frac{\partial^2 C_t}{\partial v_t^2} + \frac{1}{2} \phi^2 \eta_t \frac{\partial^2 C_t}{\partial \eta_t^2} + \rho_{12} \sqrt{v_t} \sqrt{\eta_t} S_t \frac{\partial^2 C_t}{\partial S_t \partial v_t} \\ & + \rho_{13} \phi \sqrt{v_t} \sqrt{\eta_t} S_t \frac{\partial^2 C_t}{\partial S_t \partial \eta_t} + \rho_{23} \phi \eta_t \frac{\partial^2 C_t}{\partial v_t \partial \eta_t}. \end{aligned} \quad (A1.8)$$

Meanwhile, the valuation equation that determines the price of the call option C_t is

$$r S_t \frac{\partial C_t}{\partial S_t} + (\theta(v_t) - \lambda_t^v) \frac{\partial C_t}{\partial v_t} + (\gamma(\eta_t) - \lambda_t^\eta) \frac{\partial C_t}{\partial \eta_t} + b_t - r C_t = 0. \quad (A1.9)$$

Rearranging Equation (A1.8), we obtain

$$b_t = r \left(C_t - S_t \frac{\partial C_t}{\partial S_t} \right) - (\theta(v_t) - \lambda_t^v) \frac{\partial C_t}{\partial v_t} - (\gamma(\eta_t) - \lambda_t^\eta) \frac{\partial C_t}{\partial \eta_t}. \quad (A1.10)$$

Considering that the dynamics of v_t and η_t are defined by Equations (A1.2) and (A1.3), respectively, these can be simplified by substituting b_t in Equation (A1.10) into Equation (A1.7), as follows:

$$\pi_{t,t+\tau} = \int_t^{t+\tau} \lambda_u^v \frac{\partial C_u}{\partial v_u} du + \int_t^{t+\tau} \lambda_u^\eta \frac{\partial C_u}{\partial \eta_u} du + \int_t^{t+\tau} \sqrt{\eta_u} \frac{\partial C_u}{\partial v_u} d\omega_u^2 + \int_t^{t+\tau} \phi \sqrt{\eta_u} \frac{\partial C_u}{\partial \eta_u} d\omega_u^3. \quad (\text{A1.11})$$

Therefore, the expected value of the delta-hedged gain (i.e., $\pi_{t,t+\tau}$) under physical measure P is given by considering the martingale property of Itô's integral:

$$E_t[\pi_{t,t+\tau}] = \int_t^{t+\tau} E_t \left[\lambda_u^v \frac{\partial C_u}{\partial v_u} \right] du + \int_t^{t+\tau} E_t \left[\lambda_u^\eta \frac{\partial C_u}{\partial \eta_u} \right] du, \quad (\text{A1.12})$$

where λ_t^v and λ_t^η denote compensation for stochastic volatility and VOV risks, respectively. Moreover, $\frac{\partial C_u}{\partial v_u}$ and $\frac{\partial C_u}{\partial \eta_u}$ represent the vega and volga of the call option, respectively. Equation (A1.12) shows that the expected delta-hedged gains depend on the risk premiums of volatility and VOV.

For tractability, [Huang et al. \(2019\)](#) assume that the risk premiums have a linear structure:

$$\lambda_t^v = \lambda^v v_t, \quad \lambda_t^\eta = \lambda^\eta \eta_t, \quad (\text{A1.13})$$

where λ_t^v and λ_t^η represent constant market prices of volatility and VOV risks, respectively.

The scaled expected delta-hedged gain is linear with respect to the fundamental factors by applying Itô's Taylor expansions to Equation (A1.11):

$$\frac{E_t[\pi_{t,t+\tau}]}{S_t} = \lambda^v \beta_t^v v_t + \lambda^\eta \beta_t^\eta \eta_t, \quad (\text{A1.14})$$

where β_t^v and β_t^η represent the options' exposure to volatility risk and VOV risk, respectively.

Specifically, β_t^v and β_t^η are expressed as

$$\beta_t^v = \sum_{n=0}^{\infty} \frac{\tau^{1+n}}{(1+n)!} \Phi_{t,n}^v > 0, \quad (\text{A1.15})$$

$$\beta_t^\eta = \sum_{n=0}^{\infty} \frac{\tau^{1+n}}{(1+n)!} \Phi_{t,n}^\eta > 0, \quad (\text{A1.16})$$

where $\Phi^v(\cdot)$ and $\Phi^\eta(\cdot)$ are positive functions depending on option moneyness, vega, and volga (i.e., m , $\frac{\partial C_u}{\partial v_u}$ and $\frac{\partial C_u}{\partial \eta_u}$, respectively).

A.2. Stochastic Volatility-of-Volatility Model with a Jump

Under the physical measure P , we assume that the dynamics of the underlying price S_t can be expressed as:

$$\frac{dS_t}{S_t} = \mu(S_t, v_t, \eta_t)dt + \sqrt{v_t}d\omega_t^1 + (e^x - 1)dq_t - \mu_J \Lambda_J \sqrt{v_t}dt, \quad (\text{A2.1})$$

$$dv_t = \theta(v_t)dt + \sqrt{\eta_t}d\omega_t^2, \quad (\text{A2.2})$$

$$d\eta_t = \gamma(\eta_t)dt + \phi\sqrt{\eta_t}d\omega_t^3. \quad (\text{A2.3})$$

This framework allows random jumps, along with stochastic volatility and VOV, to play a role in pricing the underlying S_t . The jump term follows a Poisson process with volatility-independent intensity $\Lambda_J v_t$. We use x to denote the jump size and q to represent its corresponding physical density. We further assume that the mean of $e^x - 1$ is μ_J , and then the compensation is $\mu_J \Lambda_J \sqrt{v_t}dt$.

Assume that, under the risk-neutral probability measure Q , the density of the jump size (i.e., x) is q , and the mean of $e^x - 1$ is μ^*_J . Then, we can write the dynamics of the underlying S_t as

$$\frac{dS_t}{S_t} = rdt + \sqrt{v_t}d\widetilde{\omega}_t^1 + (e^x - 1)dq_t^* - \mu_J^* \Lambda_J \sqrt{v_t}dt, \quad (\text{A2.4})$$

$$dv_t = (\theta(v_t) - \lambda_t^v)dt + \sqrt{\eta_t}d\widetilde{\omega}_t^2, \quad (\text{A2.5})$$

$$d\eta_t = (\gamma(\eta_t) - \lambda_t^\eta)dt + \phi\sqrt{\eta_t}d\widetilde{\omega}_t^3. \quad (\text{A2.6})$$

By Itô's lemma, we can write the call option price at time $t + \tau$ as

$$\begin{aligned} C_{t+\tau} = & C_t + \int_t^{t+\tau} \frac{\partial C_u}{\partial S_u} dS_u + \int_t^{t+\tau} \frac{\partial C_u}{\partial v_u} dv_u + \int_t^{t+\tau} \frac{\partial C_u}{\partial \eta_u} d\eta_u \\ & + \Lambda_J \int_t^{t+\tau} \sqrt{v_u} \int_{-\infty}^{\infty} [C(S_t e^x) - C(S_t)] q(x) dx du + \int_t^{t+\tau} b_u du, \end{aligned} \quad (\text{A2.7})$$

where b_t is defined as in Equation (A1.2). Moreover, the valuation equation is given by

$$\begin{aligned} (r - \mu_J^* \Lambda_J \sqrt{v_t}) S_t \frac{\partial C_t}{\partial S_t} + (\theta(v_t) - \lambda_t^v) \frac{\partial C_t}{\partial v_t} + (\gamma(\eta_t) - \lambda_t^\eta) \frac{\partial C_t}{\partial \eta_t} \\ + \Lambda_J \sqrt{v_t} \int_{-\infty}^{\infty} [C(S_t e^x) - C(S_t)] q(x)^* dx + b_t - rC_t = 0. \end{aligned} \quad (\text{A2.8})$$

Rearranging Equation (A2.8), we obtain

$$\begin{aligned} b_t = & r \left(C_t - S_t \frac{\partial C_t}{\partial S_t} \right) - (\theta(v_t) - \lambda_t^v) \frac{\partial C_t}{\partial v_t} - (\gamma(\eta_t) - \lambda_t^\eta) \frac{\partial C_t}{\partial \eta_t} \\ & + \mu_J^* \Lambda_J \sqrt{v_t} S_t \frac{\partial C_t}{\partial S_t} - \Lambda_J \sqrt{v_t} \int_{-\infty}^{\infty} [C(S_t e^x) - C(S_t)] q(x)^* dx. \end{aligned} \quad (\text{A2.9})$$

Therefore, the delta-hedged gain (i.e., $\pi_{t,t+\tau}$) under physical measure P is

$$\begin{aligned}
E_t[\pi_{t,t+\tau}] &= \int_t^{t+\tau} E_t \left[\lambda_u^v \frac{\partial C_u}{\partial v_u} \right] du + \int_t^{t+\tau} E_t \left[\lambda_u^\eta \frac{\partial C_u}{\partial \eta_u} \right] du + \mu_J^* \Lambda_J \int_t^{t+\tau} E_t \left[\sqrt{v_u} S_u \frac{\partial C_u}{\partial S_u} \right] du \\
&\quad - \Lambda_J \int_t^{t+\tau} \sqrt{v_u} \left[\int_{-\infty}^{\infty} C(S_u e^x) q(x)^* dx - \int_{-\infty}^{\infty} C(S_u e^x) q(x) dx \right] du.
\end{aligned} \tag{A2.10}$$

Unlike the expected delta-hedged gain under stochastic volatility framework in Equations (A1.1)-(A1.3), if jump fears are considered, Equation (A2.10) indicates that the delta-hedged gain is also affected by jump risk.

A.3. Construction of Risk-Neutral Moments

According to [Bakshi et al. \(2003\)](#), the prices of the volatility and cubic and quartic contracts can be expressed by option prices, respectively, as follows:

$$V_{t,t+\tau} = \int_{S_t}^{\infty} \frac{2 \left(1 - \ln \left(\frac{K}{S_t} \right) \right)}{K^2} C(t, \tau, K) dK + \int_0^{S_t} \frac{2 \left(1 + \ln \left(\frac{S_t}{K} \right) \right)}{K^2} P(t, \tau, K) dK, \tag{A3.1}$$

$$W_{t,t+\tau} = \int_{S_t}^{\infty} \frac{6 \ln \left(\frac{K}{S_t} \right) - 3 \left(\ln \left(\frac{K}{S_t} \right) \right)^2}{K^2} C(t, \tau, K) dK - \int_0^{S_t} \frac{6 \ln \left(\frac{S_t}{K} \right) + 3 \left(\ln \left(\frac{S_t}{K} \right) \right)^2}{K^2} P(t, \tau, K) dK, \tag{A3.2}$$

$$X_{t,t+\tau} = \int_{S_t}^{\infty} \frac{12 \left(\ln \left(\frac{K}{S_t} \right) \right)^2 - 4 \left(\ln \left(\frac{K}{S_t} \right) \right)^3}{K^2} C(t, \tau, K) dK + \int_0^{S_t} \frac{12 \left(\ln \left(\frac{S_t}{K} \right) \right)^2 + 4 \left(\ln \left(\frac{S_t}{K} \right) \right)^3}{K^2} P(t, \tau, K) dK, \tag{A3.3}$$

where $C(t, \tau, K)$ stands for the price of a call option with time to maturity τ and strike price K at time t . Note that, in practice, we discretized the integrals in these equations by defining the increment ΔK as

$$\Delta I(K_i) = \begin{cases} \frac{K_{i+1} - K_{i-1}}{2}, & 0 \leq i \leq N \quad (\text{with } K_{-1} = 2K_0 - K_1, K_{N+1} = 2K_N - K_{N-1}), \\ 0, & \text{otherwise.} \end{cases} \tag{A3.4}$$

A.4. Expressions in the Model-Free Jump Measure

The detailed expression for ZJ in the model-free jump measure (i.e., J) shown in Equation (21), is

$$ZJ_t = \frac{\frac{RV_t - BV_t}{RV_t}}{\sqrt{[(\pi/2)^2 + \pi - 5]/m \times \max(1, TP_t/BV_t^2)}}, \quad (\text{A4.1})$$

where

$$TP_t = m\mu_{4/3}^{-3} \frac{m}{m-2} \sum_{i=3}^m |r_{t,i-2}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i}|^{4/3}, \quad (\text{A4.2})$$

$$\mu_k = 2^{k/2} \frac{\Gamma[(k+1)/2]}{\Gamma[1/2]}, k > 0, \quad (\text{A4.3})$$

and $\Gamma(\cdot)$ denotes the gamma function.

Table 1. Descriptive statistics

	Mean	Std. dev.	Skewness	Kurtosis	Q1	Median	Q3
<i>vol</i>	33.040	10.850	0.823	3.604	25.590	31.360	39.300
<i>vov</i>	0.002	0.002	4.681	36.711	0.001	0.002	0.003
VIX	16.560	5.559	1.872	7.604	12.900	15.200	18.510
VVIX	90.670	12.912	1.146	7.050	82.340	89.180	97.570

Note: This table reports descriptive statistics such as the mean (Mean), standard deviation (Std. dev.), skewness (Skewness), kurtosis (Kurtosis), 25th percentile (Q1), median (Median), and 75th percentile (Q3) for the following key variables: crude oil market volatility (*vol*), crude oil market volatility-of-volatility (*vov*), the VIX (VIX), and the VIX volatility index (VVIX). The sample period is July 2010 to June 2018.

Table 2. Correlations

	<i>vol</i>	<i>vov</i>	VIX	VVIX
<i>vol</i>	1.000	0.250	0.493	0.427
<i>vov</i>		1.000	0.424	0.169
VIX			1.000	0.529

Note: This table reports the correlations between the following key variables: crude oil market volatility (*vol*), crude oil market VOV (*vov*), the VIX (VIX), and the VIX volatility index (VVIX). The sample period is July 2010 to June 2018.

Table 3. Delta-Hedged Gains for One-Month USO Options

Moneyness	$\pi_{t,t+\tau}$ in \$					$\frac{\pi_{t,t+\tau}(\text{in } \$)}{S_t}$ in %			
	Mean	t-stat	% < 0	Std.	AR(1)	Mean	t-stat	Std.	AR(1)
Call options									
0.900–0.925	−0.050	−2.36	59%	0.16	0.15	−0.13%	−1.86	0.01	0.05
0.925–0.950	−0.059	−2.19	61%	0.19	0.19	−0.17%	−1.99	0.01	0.06
0.950–0.975	−0.058	−1.83	60%	0.21	0.21	−0.19%	−1.96	0.01	0.15
0.975–1.000	−0.056	−1.86	59%	0.20	0.21	−0.17%	−1.65	0.01	0.12
1.000–1.025	−0.065	−2.06	64%	0.21	0.11	−0.18%	−1.83	0.01	−0.03
1.025–1.050	−0.050	−2.06	63%	0.21	−0.02	−0.15%	−2.03	0.01	−0.10
1.050–1.075	−0.058	−2.39	68%	0.20	−0.13	−0.17%	−2.19	0.01	−0.13
1.075–1.100	−0.044	−2.79	69%	0.13	0.01	−0.14%	−2.84	0.00	−0.07
Put options									
0.900–0.925	−0.068	−2.90	68%	0.18	0.13	−0.19%	−2.37	0.01	0.11
0.925–0.950	−0.076	−2.27	63%	0.23	0.22	−0.20%	−1.95	0.01	0.10
0.950–0.975	−0.070	−2.07	60%	0.23	0.13	−0.19%	−1.97	0.01	0.00
0.975–1.000	−0.071	−2.18	66%	0.21	0.27	−0.21%	−1.95	0.01	0.23
1.000–1.025	−0.069	−2.01	64%	0.23	0.17	−0.18%	−1.78	0.01	0.03
1.025–1.050	−0.060	−2.50	66%	0.19	0.06	−0.16%	−2.20	0.01	−0.07
1.050–1.075	−0.084	−3.61	70%	0.20	−0.13	−0.24%	−3.07	0.01	−0.12
1.075–1.100	−0.038	−1.85	68%	0.20	−0.12	−0.11%	−1.89	0.01	−0.15

Note: The table reports the gains on a portfolio of a long position in a one-month USO option hedged by a short position in the underlying asset, such that the net investment earns the risk-free interest rate. The sample period is July 2010 to June 2018. The delta-hedged gains are rebalanced on a daily basis and calculated based on Equation (3). We report the unscaled delta-hedged gains ($\Pi_{t,t+\tau}$) and the delta-hedged gains scaled by the underlying ($\frac{\pi_{t,t+\tau}}{S_t}$).

Both are averaged over the specified moneyness ranges, in which moneyness is defined as $m \equiv \frac{Se^{r\tau}}{K}$. The t -statistic tests the null hypothesis that the delta-hedged gains equal zero. The % < 0 column shows the percentage of negative delta-hedged gains within the corresponding moneyness range.

Table 4. Delta-Hedged Gains for One-Week USO Options

Moneyness	$\pi_{t,t+\tau}$ in \$					$\frac{\pi_{t,t+\tau}(\text{in } \$)}{S_t}$ in %			
	Mean	t -stat	% < 0	Std.	AR(1)	Mean	t -stat	Std.	AR(1)
Call options									
0.900–0.925	−0.015	−2.99	65%	0.06	0.01	−0.05%	−2.21	0.00	−0.14
0.925–0.950	−0.018	−2.96	62%	0.09	−0.05	−0.05%	−2.15	0.00	−0.09
0.950–0.975	−0.028	−3.86	63%	0.12	−0.02	−0.09%	−3.84	0.00	−0.06
0.975–1.000	−0.031	−3.46	61%	0.13	0.03	−0.09%	−2.73	0.01	0.04
1.000–1.025	−0.024	−2.72	61%	0.15	−0.07	−0.06%	−1.91	0.01	0.00
1.025–1.050	−0.014	−1.30	68%	0.17	−0.03	−0.07%	−2.09	0.01	−0.07
1.050–1.075	−0.026	−3.09	61%	0.11	0.04	−0.08%	−2.97	0.00	−0.06
1.075–1.100	−0.015	−1.58	59%	0.14	0.02	−0.08%	−2.34	0.00	0.02
Put options									
0.900–0.925	−0.012	−1.43	54%	0.12	0.02	0.00%	−0.07	0.01	0.02
0.925–0.950	−0.033	−3.62	59%	0.12	0.00	−0.10%	−3.30	0.00	−0.07
0.950–0.975	−0.034	−3.77	60%	0.12	0.10	−0.10%	−3.48	0.00	0.02
0.975–1.000	−0.033	−3.60	62%	0.13	0.02	−0.10%	−3.05	0.01	0.02
1.000–1.025	−0.028	−3.34	63%	0.14	−0.12	−0.07%	−2.13	0.01	−0.03
1.025–1.050	−0.025	−3.23	68%	0.12	0.00	−0.09%	−3.33	0.00	−0.08
1.050–1.075	−0.023	−3.49	74%	0.09	0.11	−0.08%	−3.25	0.00	0.02
1.075–1.100	−0.007	−0.87	68%	0.11	−0.02	−0.03%	−1.06	0.00	−0.03

Note: The table reports the gains on a portfolio of a long position in a one-week USO option hedged by a short position in the underlying asset, such that the net investment earns the risk-free interest rate. The sample period is July 2010 to June 2018. The delta-hedged gains are rebalanced on a daily basis and calculated based on Equation (3). We report the unscaled delta-hedged gains ($\Pi_{t,t+\tau}$) and the delta-hedged gains scaled by the underlying ($\frac{\pi_{t,t+\tau}}{S_t}$).

Both are averaged over the specified moneyness ranges, in which moneyness is defined as $m \equiv \frac{Se^{r\tau}}{K}$. The t -statistic tests the null hypothesis that the delta-hedged gains equal zero. The % < 0 column shows the percentage of negative delta-hedged gains within the corresponding moneyness range.

Table 5. Delta-Hedged Gains from Volatility and Volatility-of-Volatility Risks

	Vega		Volga	
	Slope	<i>t</i> -stat	Slope	<i>t</i> -stat
Gains _{<i>t,t+τ</i>}	−0.058***	(−4.80)	−0.002	(−0.61)
	−0.119***	(−7.22)	−0.020***	(−5.38)

Note: This table reports the cross-sectional regressions of one-month USO option gains on the vega ($\frac{\partial C_t}{\partial \sigma_t}$) and volga ($\frac{\partial^2 C_t}{\partial^2 \sigma_t^2}$) of the options, taking into account time-fixed effects. All the slopes are scaled by 1,000. The option greeks are implied by Black and Scholes (1973). The sample period is July 2010 to June 2018. The values in parentheses are Newey and West (1987) robust *t*-statistics (with lag 3), significant at the 1%, 5%, and 10% levels, denoted respectively by ***, **, and *.

Table 6. Delta-Hedged Gains from Volatility and Volatility-of-Volatility Risks

$\pi_{t,t+\tau}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
vov^2		-2.450e-06*** (-7.19)	-2.310e-06*** (-6.58)	-2.450e-06*** (-7.25)	-2.550e-06*** (-7.26)	-5.090e-06*** (-12.48)	-5.640e-06*** (-13.40)
vol^2	-0.894*** (-7.83)	-0.610*** (-5.17)	-0.553*** (-4.83)	-0.548*** (-4.65)	-0.578*** (-4.77)	-0.410*** (-3.64)	-0.313*** (-2.76)
$skew$			3.201e-04* (1.87)		-2.28E-04 (-1.08)		
$kurt$				-2.751e-05*** (-4.58)	-3.237e-05*** (-4.31)		
J						1.037*** (10.48)	
J_+							0.623*** (4.36)
J_-							1.390*** (11.24)
$\pi_{t-\tau,t}$	0.06 (1.87)	0.031 (0.96)	0.033 (1.01)	0.047 (1.47)	0.047 (1.45)	0.070** (2.31)	0.049 (1.63)

Note: This table shows the predictability of one-month-ahead delta-hedged gains from volatility (vol), VOV (vov), and jump (i.e., $skew$, $kurt$, J , J^+ , and J^-) risks. The sample period is July 2010 to June 2018. The values in parentheses are Newey and West (1987) robust t -statistics (with lag 3), significant at the 1%, 5%, and 10% levels, denoted respectively by ***, **, and *.

Table 7. Information content of oil volatility-of-volatility on macro variables

Macro Variables	vov^2	vol^2	VVIX ²	AR1	TS	DS	DP	Adj. R^2 (%)
$\Delta \ln$ RPCE	-0.107** (-2.11)	5.766e-07* (1.92)	-8.290e-08 (-0.80)	-0.226** (-2.51)	-0.020 (-1.50)	-0.251** (-2.27)	-0.001 (-0.33)	5.96
KCFSI	20.337*** (2.58)	7.421e-05 (1.53)	3.155e-05*** (4.06)	0.564*** (4.24)	2.135 (1.24)	19.195 (0.75)	0.195 (0.54)	69.78

Note: This table reports the one-month ahead predictability of oil VOV (vov) on the monthly growth rate of real personal consumption ($\Delta \ln$ RPCE) and Kansas City Financial Stress Index (KCFSI), controlling for oil volatility (vol), equity VOV (VVIX), the term spread (TS), the default spread (DS), and the log dividend–price ratio (DP) and its lag term (AR1). The sample period is July 2010 to June 2018. The values in parentheses are Newey and West (1987) robust t -statistics (with lag 3), significant at the 1%, 5%, and 10% levels, denoted respectively by ***, **, and *.

Table 8. Robustness check: Another volatility-of-volatility measure

$\pi_{t,t+\tau}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
vov^2		-2.770*** (-8.31)	-2.570*** (-7.41)	-2.740*** (-8.28)	-2.550e-06*** (-7.26)	-5.380*** (-13.70)	-5.830*** (-14.64)
vol^2	-0.122*** (-5.25)	-0.075*** (-3.26)	-0.061** (-2.55)	-0.066*** (-2.86)	-0.578*** (-4.77)	-0.056*** (-2.60)	-0.060*** (-2.80)
$skew$			41.404** (2.41)		-2.28E-04 (-1.08)		
$kurt$				-2.948*** (-4.89)	-3.237e-05*** (-4.31)		
J						1.077*** (10.98)	
J_+							0.581*** (4.05)
J_-							1.458*** (12.16)
$\pi_{t-\tau,t}$	0.095*** (2.76)	0.048 (1.41)	0.047 (1.38)	0.063* (1.87)	0.064* (1.89)	0.086** (2.70)	0.067*** (2.13)

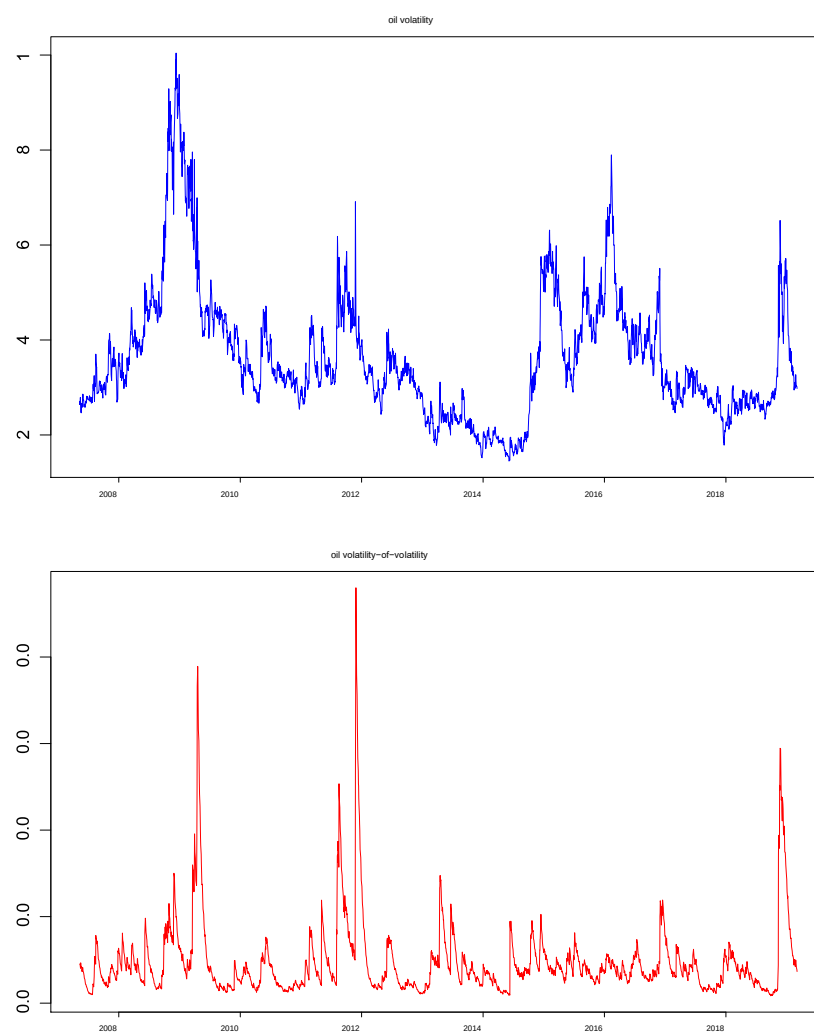
Note: This table shows the predictability of one-month-ahead delta-hedged gains from volatility (vol), volatility of-volatility (vov^B), and jump (i.e., $skew$, $kurt$, J , J^+ , and J^-) risks. The sample period is July 2010 to June 2018. The values in parentheses are Newey and West (1987) robust t -statistics (with lag 3), significant at the 1%, 5%, and 10% levels, denoted respectively by ***, **, and *.

Table 9. Robustness check: Predictability of one-week delta-hedge gains

$\pi_{t,t+\tau}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
vov^2		-8.817e-07*** (-7.43)	-9.915e-07*** (-8.01)	-9.921e-07*** (-8.04)	-1.030e-06*** (-8.26)	-1.09e-07 (-0.26)	-7.01e-08 (-0.17)
vol^2	-0.218*** (-5.36)	-0.109** (-2.53)	-0.077 (-1.60)	-0.081* (-1.68)	-0.083* (-1.72)	-0.403*** (-4.82)	-0.401*** (-4.86)
$skew$			-1.38E-04 (-0.93)		-3.523e-04** (-1.99)		
$kurt$				-2.12E-05 (-1.37)	-4.117e-05** (-2.23)		
J						0.337*** (4.00)	
J_+							0.596*** (5.60)
J_-							0.129 (1.25)
$\pi_{t-\tau,t}$	-0.024 (-1.25)	0.025 (-1.31)	0.024 (-1.12)	-0.026 (-1.32)	-0.025 (-1.27)	-0.026* (-0.63)	-0.037 (-0.88)

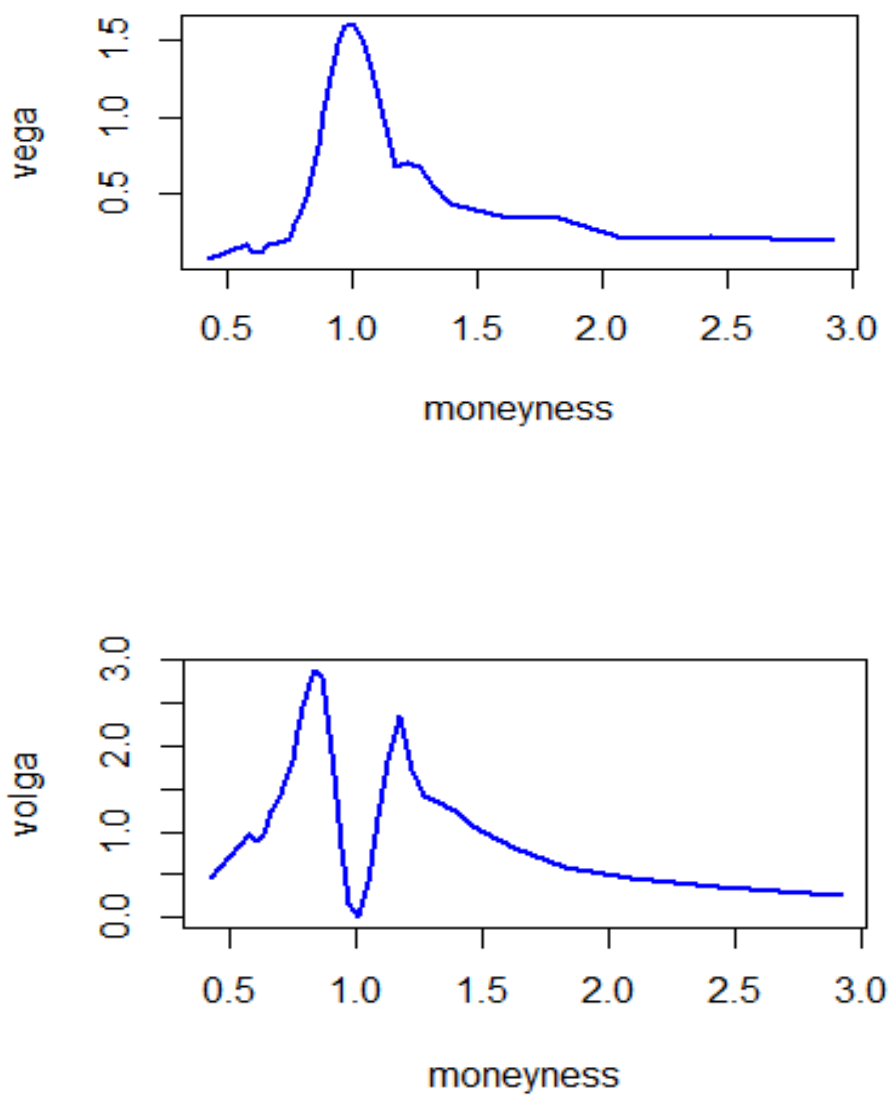
Note: This table shows the predictability of one-week-ahead delta-hedged gains from volatility (vol), volatility of-volatility (vov), and jump (i.e., $skew$, $kurt$, J , J^+ , and J^-) risks. The sample period is July 2010 to June 2018. The values in parentheses are Newey and West (1987) robust t -statistics (with lag 3), significant at the 1%, 5%, and 10% levels, denoted respectively by ***, **, and *.

Figure 1: Oil market volatility (OVX) and volatility-of-volatility (VOV)



Note: This figure shows the time series of OVX (upper panel) and oil VOV (lower panel). The sample period is July 2010 to June 2018.

Figure 2: Call vega and volga



Note: This figure plots the Black–Scholes–implied vega (i.e., $\frac{\partial C_t}{\partial \sigma_t}$, upper panel) and volga (i.e., $\frac{\partial^2 C_t}{\partial^2 \sigma_t^2}$, lower panel) for USO call options on September 18, 2015, with time to maturity of one month with respect to their moneyness (i.e., $m = Se^{r\tau}/K$).